

❖ **RECIPE PREPARATION AGENT USING AI & MACHINE LEARNING**

IBM University Engagement Project
Submission

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***Domain: Artificial Intelligence & Machine
Learning***

Project Tile – Recipe Preparation Agent

Domain of Project – **Artificial Intelligence & Machine Learning**

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Introduction

Introduction to AI & Machine Learning

Artificial Intelligence (AI) enables machines to simulate human intelligence.

Machine Learning (ML) is a subset of AI that allows systems to learn from data and make predictions without explicit programming.

➤ Application of AI in Food Industry:-

- Smart Recipe Recommendation
- Food Analysis
- Nutrition Planning
- Personalized Cooking Assistance
- Restaurant Automation

Revolutionary Use Cases of AI in the Food Industry



➤ Problem Statement:-

People often struggle to decide what food can be prepared using available ingredients.

Searching recipes manually is time-consuming and inefficient.

➤ Objective:-

Develop an AI-powered Recipe Preparation Agent that recommends recipes based on ingredients entered by the user.



Objectives:-

Project Objectives:

Analyze recipe dataset

Search recipes using ingredients

Display recipe details

Provide cooking instructions

Improve user cooking experience

Build an interactive AI-based application

Dataset Used:- Food.com Recipe Dataset

Dataset Description Dataset Information

Dataset Contains

- Recipe Name
- Ingredients
- Cooking Steps
- Number of
Ingredients
- Preparation
Time
- Nutrition
Information

Total Records:-

230,000+ Recipes
12 Dataset
Features

❖ Tools & Technologies Used:-

Technologies Used

Programming Language:-

Python

Development Environment:-

Jupyter Notebook

VS Code

Libraries:-

Pandas

NumPy

Matplotlib

Streamlit

Version Control:-

Git

GitHub

System Workflow

Recipe Recommendation Workflow

Workflow

Data Collection



Data Cleaning



Recipe Dataset Processing



Ingredient Search

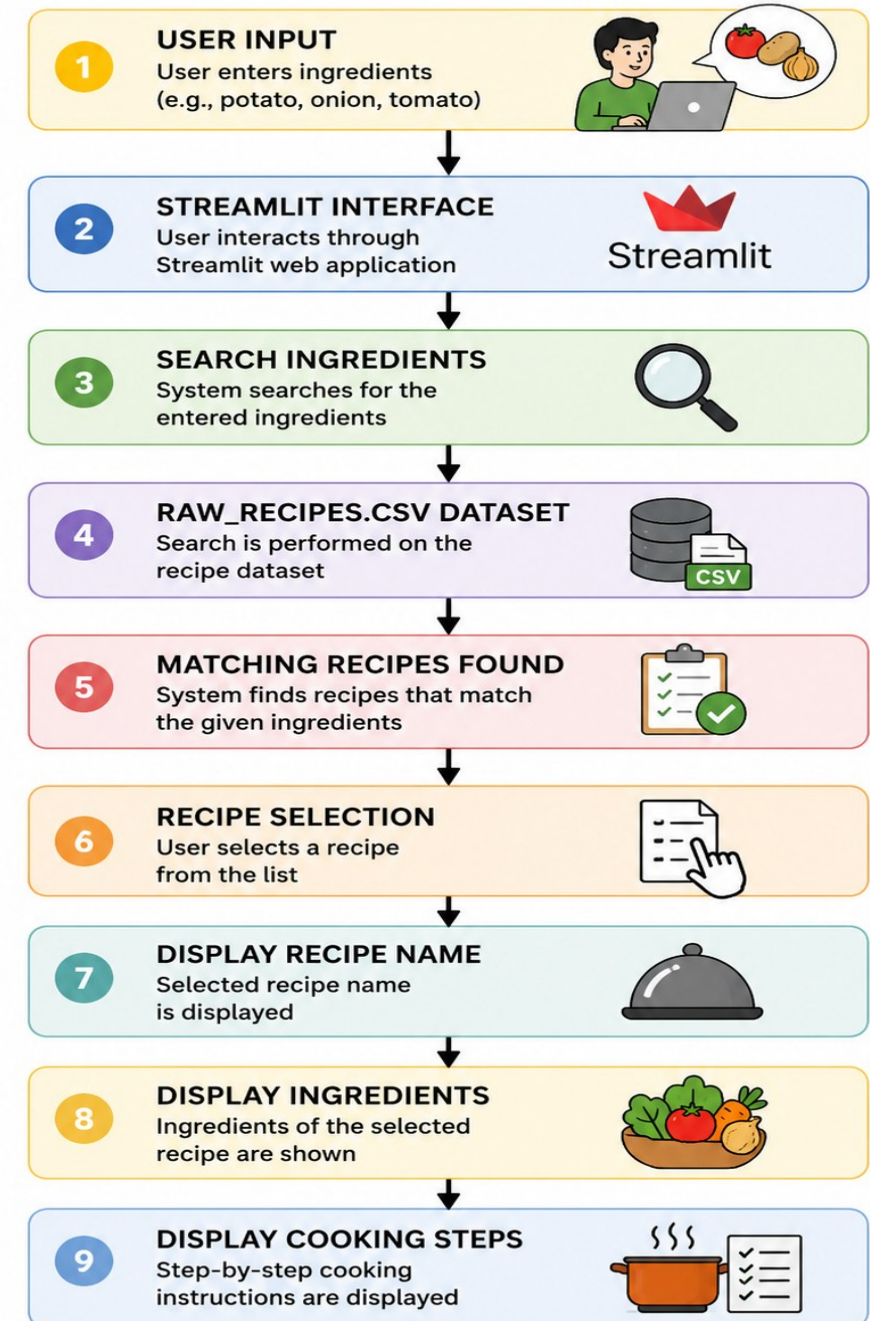


Recipe Recommendation



Display Recipe Details

RECIPE PREPARATION AGENT WORKFLOW



Data Visualization & Graph

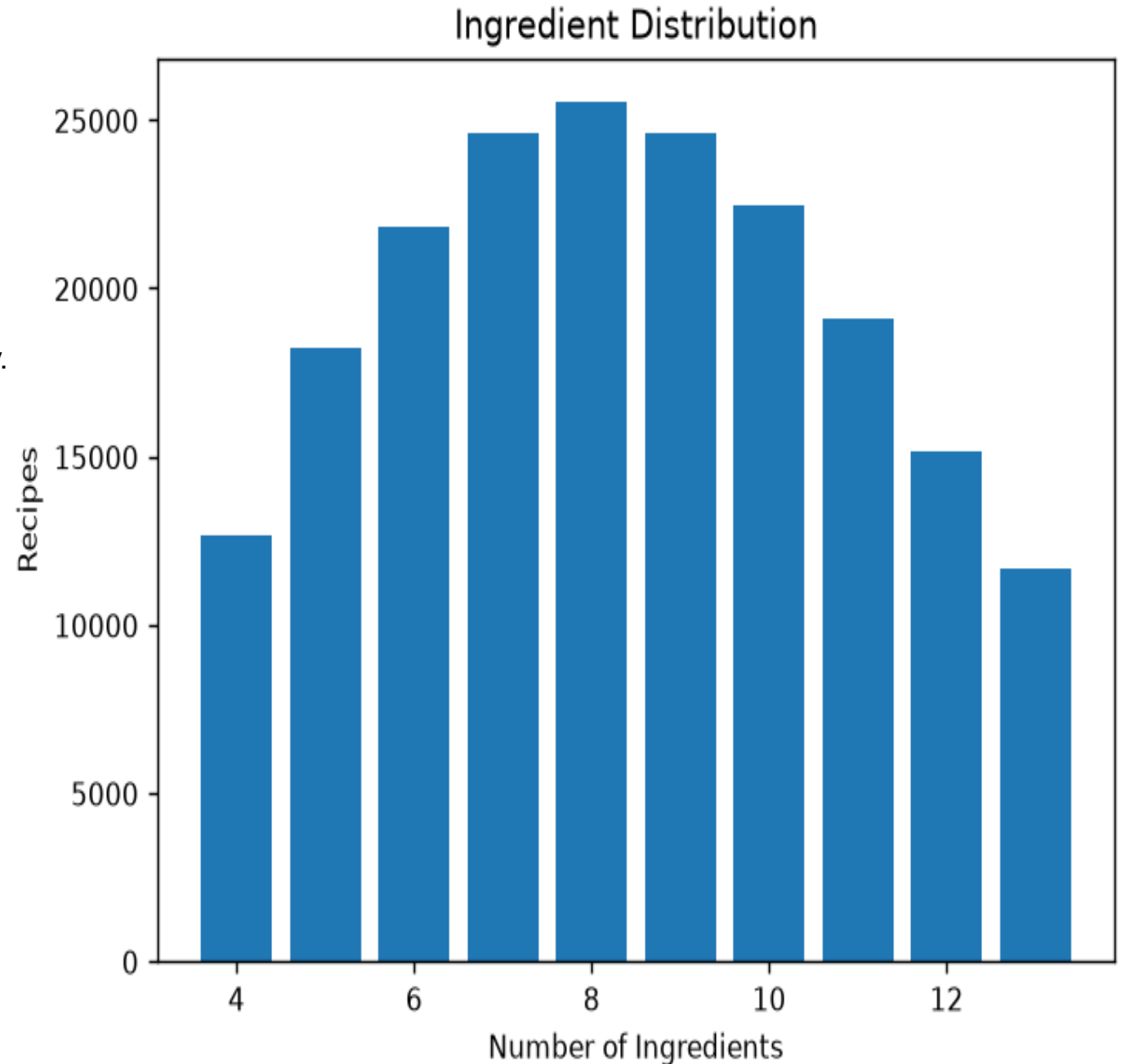
➤ Data Visualization:

❖ Recipe Distribution by Number of Ingredients

➤ Observations:-

- Most recipes contain 5–10 ingredients.
- Dataset contains a large variety of recipes.
- Ingredient count significantly impacts recipe complexity.

• Graph:-



Results:-

Successfully Implemented Features

- Ingredient-based Search
- Recipe Recommendation
- Recipe Details Display
- Cooking Instructions
- Interactive User Interface

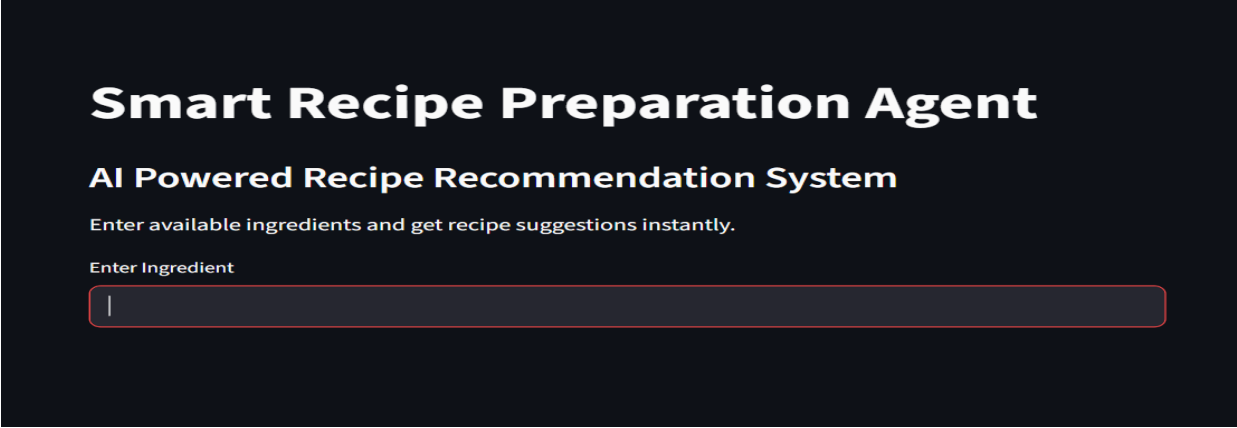
Output Example:-

Input:-

potato

Output:

- Potato Soup
- Potato Curry
- Garlic Potato Bake



Benefits

Saves Time

Easy Recipe Discovery

User Friendly

AI-Based Recommendation

Better Cooking Experience

Future Scope:-

- Voice Assistant Integration
- Mobile Application
- Nutrition Recommendation
- AI Chatbot Integration
- Personalized Recipe Suggestions

➤ Git Hub Link:-

Github URL :-

<https://github.com/codebydineshkumar>

Another Repositories:-

<https://github.com/codebydineshkumar/Task-1-EDA>

<https://github.com/codebydineshkumar/AI-ML-Task2-DataCleaning>

https://github.com/codebydineshkumar/Task3_EDA

<https://github.com/codebydineshkumar/Task-4-FES>

<https://github.com/codebydineshkumar/Task-5-Heart-Disease>

<https://github.com/codebydineshkumar/Task6-Preprocessing>

<https://github.com/codebydineshkumar/Task7-EDA>

Thank You!

Thank you for your time and interest.